Weight-Skew Erosion in Federated Retriever Fine-Tuning: Measurement, Mechanism, and Conflict-Aware Repair

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ABSTRACT

Organizations that cannot pool documents increasingly want to share retrieval competence: one dense retriever, adapted federatedly with parameter-efficient updates, serving retrieval-augmented generation over private, evolving corpora. In a search current to August 23, 2026, we found no prior study combining per-client and per-depth ranking analysis with federated, adapter-only, replay-free continual dense retrieval; this is a bounded "to our knowledge" claim, not a verified absence. Across the measured work-regime × weighting grid, the capped uniform/n_k/corpus cells have three seeds while capped max–min and every full-work cell have seed 42 only. Uniform aggregation improves every client in the measured cells, whereas size-proportional FedAvg erodes under-weighted clients near the top of the ranking and is Pareto-dominated by uniform even for the largest client. With full local epochs, seed-42 size weighting drives the two smallest clients *below the untrained backbone* (SciFact −0.070 and ArguAna −0.025 frozen-anchored) while the two largest still gain — a transient round-1 spike fully surrendered by round 8. A pre-registered mechanism discriminator adjudicates the cause: repeating the same n_k weighting in a shared frozen-A coordinate — which removes the factor-aggregation pathway by construction — *eliminates the absolute harm* at both completed seeds (SciFact +0.051/+0.055/+0.048, ArguAna +0.011/+0.019/+0.010 against the backbone, all three registered seeds clean, so the pre-registered withdrawal executed unanimously), while within that coordinate n_k remains the worst weighting for the minorities. The destruction is therefore an *interaction*: weighting skew alone redistributes; weighting skew compounded with joint two-factor training destroys, and either exact aggregation (freezing A) or de-skewed weighting removes the absolute component. A budget-matched local-only control sharpens the stakes: under n_k, every measured silo finishes below what it would have reached training alone, while uniform federation beats the equal-budget outside option for only some silos — federation is not automatically worth joining. A per-round cosine-Gram max–min weighting intervention preserves the entire round-1 gain instead of surrendering it (a +0.091 frozen-anchored swing in that cell, robust to the statistic choice), while uniform remains the score co-champion. These historical runs trained both LoRA factors and applied unit-direction game weights to raw factor states. They therefore motivate, but do not yet validate, the submitted frozen-A, norm-consistent FedSpan method; corrected implementation, attribution reruns, external baselines, and multi-session evidence are required before the method claim is submission-ready.

CCS CONCEPTS

• **Information systems** → **Information retrieval**; • **Computing methodologies** → *Machine learning*.

KEYWORDS

federated learning, dense retrieval, retrieval-augmented generation, catastrophic forgetting, LoRA, aggregation fairness

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1 INTRODUCTION

Retrieval-augmented generation (RAG) has made the dense retriever shared infrastructure: the component that decides which evidence an agent, an assistant, or a pipeline ever sees. Many agent-memory systems absorb change non-parametrically through evolving stores, graphs, and consolidation loops, but some deployments also need encoder adaptation. Organizations such as hospitals, banks, and law firms may be unable to pool the underlying documents. Federated fine-tuning of parameter-efficient adapters is one possible middle ground: share LoRA deltas, not documents.

This paper asks what happens to each organization's ranking when those adapters are trained jointly. The closest literature is fragmented: named federated-RAG systems often federate inference or use static/end-point evaluation; named continual dense-retrieval systems such as L2R and QDC are centralized; and federated fairness/common-descent methods optimize differentiable client objectives rather than per-depth held-out ranking. Our dated search found no exact match for the combined setting. We therefore test the narrower question directly rather than claiming that every adjacent problem was previously unmeasured.

In the measured grid, the weighting determines who pays. Under uniform aggregation every client improves, including the seed-42 full-epoch cell. Under size-proportional weights, minority nDCG@10 peaks and then erodes more strongly than deeper recall, and uniform Pareto-dominates it for every client including the one holding 87% of the weight. With full local epochs, the seed-42 size-weighted cell destroys minority competence in absolute terms: SciFact and ArguAna finish *below the untrained backbone* (−0.070 and −0.025 frozen-anchored) while the two large clients still gain, and SciFact surrenders a 0.150 nDCG@10 round-1 spike. The interaction is measured; its causal optimization explanation remains an experimentally testable hypothesis.

Our contributions:

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(1) **Phenomenon** (§6): a per-client, per-depth characterization of federated retriever fine-tuning over a work-regime × weighting grid. The capped uniform/n_k/corpus cells have three seeds; capped max–min and all full-work cells are explicitly seed-42-only. Findings are depth-attenuated minority erosion under skewed weights, Pareto dominance of uniform in this testbed, and a seed-42 work × skew interaction.
(2) **Theory** (§4): an absolute margin-censored nDCG@10 perturbation bound in score space, a strict two-threshold depth window, and exact fixed-factor aggregation algebra. These results do not imply signed held-out ranking improvement from client-update alignment.
(3) **Mechanism** (§7): state-based evidence that size weighting captures the linear aggregate direction and that minority directional payoff becomes negative, alongside the full time-varying bilinear residual. These are descriptive mechanism diagnostics, not a causal loss-to-ranking theorem.
(4) **Intervention and method candidate** (§8): historical ordinary-LoRA max–min weighting results plus a corrected frozen-A, orthonormal, norm-consistent FedSpan specification. The former are evidence for the intervention; the latter requires clean attribution reruns before it is an empirical contribution.

2 RELATED WORK

Continual and lifelong retrieval. Named centralized systems include L2R [2], FlowRAG [30], CREAM [23], and QDC [10], which directly studies continual dense-retrieval drift and forgetting. They do not instantiate our federated C1–C3 setting, but QDC prevents a broad "first ranking-level continual retrieval" claim.

Federated RAG.. A mapping study catalogs federated-RAG designs and identifies heterogeneity and evaluation gaps [3]. FedE4RAG trains BGE retrievers with communicated model updates and specifies no adapter restriction [18]. Neither supplies our adapter-only, per-client/per-depth, replay-free multi-session setting; our multi-session validation remains pending.

Federated LoRA aggregation. Fixed-factor FFA-LoRA [24], residual-corrected FedEx-LoRA [22], and stacked FLoRA [27] are direct construction ancestors. Our historical rescue did not freeze a factor, so it cannot be relabeled as FFA-LoRA; corrected attribution is an open experiment.

Optimization, fairness, and ranking robustness. MGDA and its multi-task/federated descendants [5, 12, 21], CAGrad [16], FairGrad [1], FAMO [15], regularized maximin-correlation analysis [13], AdaFed [11], and CRAFT [26] already cover minimum-norm, common-descent, or worst-improvement geometry. AFL adversarially optimizes a worst-case client-mixture risk [19]; q-FFL instead defines a q-fair transformed objective and its q-FedAvg solver [14]. Goren et al., Wu et al., and Devic et al. study ranking robustness under document, word-substitution, or prediction perturbations [4, 9, 28]. Our claim is a retrieval-specific constrained composition and diagnosis, not invention of minimax weighting or ranking robustness.

Non-parametric memory. External-store and graph-memory systems can absorb corpus evolution without cross-silo encoder training. They are adjacent deployment choices, not evidence that every deployment freezes its encoder. Our setting concerns organizations that decide encoder adaptation is necessary but cannot centralize data.

3 PROBLEM SETTING

K clients hold private, evolving corpora $C_k(t)$ and query–document supervision $P_k(t)$. They jointly maintain one retriever: a frozen backbone f_{θ_0} with LoRA adapters $\Delta W_\ell = \sigma_\ell B_\ell^{\mathrm{raw}} A_\ell$, where $\sigma_\ell = \alpha_\ell / r_\ell$ is the known PEFT scale. Constraints: **(C1)** raw data never leaves silos — only adapter parameters and $O(1)$ scalars are communicated; **(C2)** no data replay across sessions — only $o(|\mathrm{data}|)$ statistics persist; **(C3)** one global model — no personalization. Each round, every client trains its local adapter copy on its own pairs with contrastive InfoNCE and in-batch negatives. We explicitly compare capped local steps with full-local-epoch work; we do not imply one fixed-work convention for every arm. The historical runs aggregate both factors, while the candidate implementation shares and freezes A and aggregates scaled B-deltas; its clean E0 attribution remains pending. The deployment-default weighting is $w_k \propto n_k$ (local pair count); we study {uniform, n_k, corpus-size, and conflict-aware w^*}.

Instruments. After every round we evaluate every client's ranking on its own held-out queries at multiple depths (nDCG@10, recall@10, recall@100). For federation rounds the *primary cross-arm statistic is frozen-anchored drift* (final minus the untrained backbone, which we verified to be arm- and seed-independent in every measured cell), with final scores reported alongside in every cross-arm table; per-client peak-minus-final erosion E; and depth structure. Round-1-anchored drift Δ_{round} (final minus round 1) is reported only as a *within-arm* trajectory statistic: round 1 is post-treatment, and we measured its anchor differing across arms by 4.7–96 seed-sd, which invalidates it for cross-arm comparison (worked example: the corpus arm's apparent deficit is 76–85% anchor artifact, with a sign reversal against n_k on recall metrics). Sequential and future session experiments reserve GEM-style BWT/Forget terminology for genuine task/session matrices.

4 THEORY

We state the two results that organize the empirical sections; proofs, the aggregation algebra, and five further results appear in the appendix.

Margin-censored perturbation bound. For unit-normalized embeddings with per-checkpoint drift budgets $\varepsilon_q, \varepsilon_D$ and $\rho = \varepsilon_q + \varepsilon_D$: no document pair with score margin above 2ρ can invert, and per query

$$|\Delta \mathrm{nDCG}@k| \;\le\; \frac{L_k}{\mathrm{IDCG}_k} \sum_{i<j} |g_i - g_j| \, \mathbf{1}\{m_{ij} \le 2\rho\},$$

with L_k the largest adjacent discount gap. Damage is *censored by the margin profile*: only tightly separated pairs are at risk.

Depth window. Let α_k^{adj} be the smallest adjacent score gap at ranks $r, r{+}1$ with $r \le k$ and unequal gains. Under the strict sufficient score-space window $\alpha_k^{\mathrm{adj}} < 2\rho < \beta_K$ (the coverage-boundary margin floor at depth K), there exist score perturbations within the budget that change nDCG@k, while recall@K cannot decrease. Equality at $\alpha_k^{\mathrm{adj}} = 2\rho$ may create only a tie and depends on tie-breaking; embedding-space realizability needs an additional encoder-geometry argument. Empirical margin profiles, rather than this existence theorem alone, decide whether the depth-attenuation mechanism applies.

Convergence of the frozen-span family. Write $f_k(B) := L_k(W_0 + \sigma BA^*)$; if each L_k is L-smooth then f_k is L_B-smooth with $L_B = \sigma^2 c^2 L$ under $A^* A^{*\top} = c^2 I$ (transfer lemma, appendix). Consider the one-local-step, full-participation, exact-gradient instantiation: per round the server applies $B_{t+1} = B_t + \eta z_t$, where z_t is the minimum-norm point of $\mathrm{conv}\{-\nabla f_1, \ldots, -\nabla f_K\}$ and $\eta \in (0, 2/L_B)$ is constant. Then, with $c_\eta = \eta(1 - L_B \eta/2) > 0$: (i) *every* client's loss decreases monotonically, $f_k(B_{t+1}) \le f_k(B_t) - c_\eta \|z_t\|^2$ for all k simultaneously (by the minimum-norm variational inequality $\langle -\nabla f_k, z_t \rangle \ge \|z_t\|^2$ and the descent lemma); (ii) if at least one f_k is bounded below, $\sum_t \|z_t\|^2 \le D/c_\eta$ with $D = \min_k (f_k(B_0) - f_k^{\inf})$ over the bounded-below clients, hence $\min_{t<T} \|z_t\|^2 = O(1/T)$; (iii) every accumulation point $\bar{B}$ is Pareto-stationary, $0 \in \mathrm{conv}\{\nabla f_k(\bar{B})\}$, since $B \mapsto \mathrm{dist}(0, \mathrm{conv}\{\nabla f_k(B)\})$ is continuous (the Hausdorff distance between the hulls is at most $\max_k \|\nabla f_k(B) - \nabla f_k(B')\|$) and equals $\|z_t\| \to 0$ along the iterates. *Attribution:* this is a specialization of classical multiobjective steepest descent to the federated frozen-span coordinate — the min-norm common-descent direction of Fliege & Svaiter and Désidéri (whose projection inequality supplies the sharp factor) [5, 7], the constant-stepsize $O(1/T)$ criticality-rate template of Fliege, Vaz & Vicente [8] (see also Tanabe et al. [25]), whose federated one-local-step form coincides with the exact-gradient case of FMGDA [29]; our additions are the frozen-span constant $L_B = \sigma^2 c^2 L$ and the explicit per-client constant c_η. Proof in the appendix, included for completeness.

Two boundary statements keep the theorem honest. First, it covers the *ungated* iteration on all K clients: the deployed activity gate (which excludes zero or non-finite updates) is a separate mechanism, and a gated-out client with a tiny nonzero gradient is not covered by (i) — it can ascend by at most $\eta \|\nabla f_k\| \|z_t\| + (L_B/2)\eta^2 \|z_t\|^2$ in that round, a quantity the gate threshold bounds. Second, the *deployed* method applies the normalized direction $d_t^\star = z_t / \|z_t\|$ at the median client-update norm, which lies outside this theorem class *by design*: the guarantee-bearing variant with step $\eta_t \gamma_t^\star$ instantiates FedMGDA+ at ε=1 [12] (FedMDFG [20] and AdaFed [11] develop the adjacent fair common-descent line) — inheriting its Theorem 1b modulo assumptions multi-step LoRA deltas do not verifiably satisfy (globally bounded gradients; deterministic plain-GD local steps with vanishing local rate; vanishing weight deviation) — at the price of a step that shrinks with the conflict level, the stall our step policy removes. We note defensively that FedMGDA+'s own theorems likewise analyze the unnormalized iteration, so the corollary inherits at exactly the strength their theory covers their deployed algorithm. On a two-client toy quadratic the $\eta_t \gamma_t^\star$ variant converges to the Pareto set while the constant-norm step locks into a bounded orbit around it; on the real grid, the measured $\gamma_t^\star \in [0.457, 0.644]$ with zero fallback rounds indicates the orbit regime, not the collapse regime, is what deployment trades for stall-freedom.

Aggregation algebra. Factor averaging deviates from matrix-space averaging by exactly the factor cross-covariance $\sum_k w_k (B_k - \bar{B})(A_k - \bar{A})$ — zero when a factor is shared (sufficient, not necessary). With a frozen shared A^* (orthonormal rows), aggregation is exact and linear in B; the Frobenius Gram matrix of communicated B-updates equals that of the true weight-space updates, which is what makes the server-side program of §8 geometry-faithful while seeing only adapters.

5 EXPERIMENTAL SETUP

Four BEIR domains as clients — NFCorpus, FiQA-2018, SciFact, ArguAna ($n_k \approx$ 110k/14k/0.9k/0.7k pairs; corpora 3.6k/57k/5k/8.7k docs) — proxying medical/financial/scientific/argumentation silos. Backbone: Contriever (110M), selected by a headroom gate (independent fine-tuning must beat frozen on every domain; the gate rejects cells in which the chosen recipe lacks headroom — our pilot's bge-m3 failed it on two of four domains). LoRA rank 16 on attention and FFN modules; MTEB-convention self-retrieval exclusion; three seeds for capped uniform/n_k/corpus cells; capped max–min and all full-work cells are seed-42-only; per-round full re-encoding and evaluation; dirty commit-labeled historical result files. The work-regime grid: *capped* (500 local steps/round, R=15) and *saturation* (full local epochs, R=8). We report the pre-registered protocol (narrative branches and the saturation prediction were written before the data) in the appendix.

6 THE PHENOMENON

6.1 Sequential baseline: mild, order-governed

Sequentially fine-tuning one Contriever model through the four domains (three seeded orders) yields signed BWT of −0.025, −0.003, +0.001 nDCG@10 — real but mild in this measured cell, with damage concentrated when the largest domain trains last; peak-minus-final Forget is positive (+0.048/+0.045/+0.023). The separate bge-m3 pilot lacked headroom on two domains, so we make no cross-backbone claim about the severity of sequential forgetting.

6.2 Federated: the weighting decides who pays

Table 1 shows the capped regime. Under uniform weighting all clients finish above the untrained backbone (frozen-anchored drift +0.113 over three seeds; within-arm round-1-anchored drift +0.072–+0.074) and peak-to-final erosion is small but nonzero (maximum 0.00591). Under n_k, the under-weighted SciFact peaks at round 3 in *all three seeds* and erodes (E = +0.025/+0.033/+0.025) while its recall@100 stays near-flat; under corpus-size weighting the under-weighted NFCorpus erodes in *both* precision and recall ($E_{\mathrm{recall@100}}$ = +0.011/+0.013/+0.011 across seeds) — on the one eroding client far from recall ceiling, recall falls nearly proportionally, which is why we name the structure *depth-attenuated erosion* rather than a dissociation. Final scores place corpus much closer to uniform (0.457 vs. 0.468 mean nDCG@10) than the within-arm drift column suggests; corpus finishes ahead of n_k on recall at all three depths on all three seeds. Erosion is forgone federation gain, not net harm versus going alone (eroded clients still beat their independent baselines; the single exception, FiQA under n_k, never reaches its independent

Table 1: Capped regime (R=15, 500 steps/round), nDCG@10: frozen-anchored drift (primary, seed mean; n=3) and mean final score, with the within-arm round-1-anchored drift Δ_{round} per seed shown for trajectory reference, plus the replicated minority-erosion signature.

Weighting	frozen-drift	final nDCG@10	Δ_{round} (s42/s123/s2024)
uniform	+0.113	0.468	+.073/+.074/+.072
n_k	+0.067	0.423	+.018/+.021/+.019
corpus	+0.102	0.457	+.025/+.024/+.024

Table 2: The work-regime × weighting grid (seed 42, single-seed): frozen-anchored nDCG@10 drift (primary) with mean final score; the within-arm round-1-anchored Δ_{round} is shown for trajectory reference. The max–min column is the historical ordinary-LoRA intervention, not frozen-A FedSpan.

	uniform		n_k		max–min w^*	
	frozen	final	frozen	final	frozen	final
capped	+0.113	0.468	+0.066	0.421	+0.112	0.467
full epochs	+0.102	0.457	**+0.003**	0.359	+0.094	0.449

baseline and we report it). Critically, in these cells uniform beats n_k at round 15 for *every* client, including the 87%-weight NFCorpus.

6.3 The interaction: work × skew

Table 2 completes the grid. Saturation alone is positive under uniform weighting (frozen-anchored +0.102). Skew alone is damaging but bounded. Together, in the single-seed saturation cell, size weighting produces complete give-back: after a transient round-1 spike (within-arm $\Delta_{\text{round}} = -0.053$), every client falls below its round-1 value by round 8, and the destruction is absolute for the minorities — SciFact finishes 0.070 and ArguAna 0.025 *below the untrained backbone* while the two large clients still gain (+0.045, +0.064; mean frozen-anchored drift +0.003). SciFact surrenders a 0.150 nDCG@10 round-1 peak. The interaction was a registered prediction, and the per-client absolute-harm reading now *replicates on a second seed*: at seed 123 the same cell drives SciFact 0.077 and ArguAna 0.017 below the untrained backbone while NFCorpus and FiQA gain (+0.045, +0.065; mean frozen-anchored drift +0.004). A third seed is in flight.

The replication also supplies the control that makes the attribution causal. Holding seed, data, rounds and local work fixed and changing *only* the weighting rule to uniform, every client finishes above the untrained backbone and the two minorities gain the most (SciFact +0.098, ArguAna +0.100; mean +0.103). Full-epoch training therefore does not harm small clients per se: the harm is attributable to size-proportional weighting. Uniform also dominates n_k for *every* client at this horizon, including the 87%-weight NFCorpus (+0.025), replicating the lose–lose structure. Table 3 reports the cell.

Table 3: Replication and control at the saturation horizon (R=8, full epochs), seed 123, frozen-anchored nDCG@10 drift per client. Changing only the weighting rule removes the absolute harm; the two smallest clients are the ones it is removed from.

Client (train pairs)	frozen	n_k	uniform	uniform − n_k
NFCorpus (110k)	[illegible]	+0.045	+0.071	+0.025
FiQA (14k)	[illegible]	+0.065	+0.142	+0.077
SciFact (0.9k)	0.571	−0.077	+0.098	+0.175
ArguAna (0.7k)	0.454	**−0.017**	+0.100	+0.117
mean	—	+0.004	+0.103	+0.099

7 MECHANISM

Two instruments dissect the failure: per-round update geometry, and a pre-registered intervention that removes one candidate cause by construction.

Geometry: what weighting does to directions. From per-round adapter states we compute the cosine Gram of effective LoRA updates. The honest per-client statistic is *leave-one-out* alignment $\cos(\Delta W_k, \sum_{j\neq k} w_j \Delta W_j)$ — the self-inclusive alignment of the dominant client is a closed-form function of the weights and norms (it reads 0.98 under the true geometry, under an identity Gram, and under an adversarial Gram alike at $w_{\max} = 0.875$) and therefore carries no information about geometry. Under n_k at the saturation horizon, every client's leave-one-out alignment decays toward orthogonality or below (majority −0.14, SciFact −0.05 by round 15, seed 42): each participant is increasingly trained by an aggregate that the *rest* of the federation points away from. This is why the failure is lose–lose rather than a transfer. The bilinear residual between the linear-mixture proxy and factor averaging is 0.071 in weighted round 1, falling to $\sim 10^{-3}$ late; per-round alignment/next-round-nDCG@10 correlations are mixed across clients and arms and are reported as descriptive only.

Adjudication by intervention, not magnitude. We had initially read the late 10^{-3} residual as evidence that factor-aggregation error is not causal. The pre-registered discriminator refuted that inference: rerunning the *same* n_k weighting with a shared frozen A — identical skew, factor-aggregation pathway removed by construction, canonical-validator-checked — eliminates the absolute harm at all three registered seeds (SciFact +0.051/+0.055/+0.048, ArguAna +0.011/+0.019/+0.010 against the backbone, versus −0.070/−0.077/−0.07 and −0.025/−0.017/−0.015 with both factors training). A small per-round residual is not a small trajectory effect. Within the frozen coordinate the weighting ordering survives — n_k is still the worst arm for the minorities (ArguAna +0.012 under n_k versus +0.057 uniform and +0.110 FedSpan at seed 123) — so the corrected mechanism is an *interaction*: weighting skew alone redistributes; weighting skew compounded with joint two-factor local training destroys. Either removing the factor pathway (exact aggregation on a frozen span) or removing the skew (uniform or geometry-aware weights) suffices to remove the absolute component. This places the factor-aggregation repair literature (FFA-LoRA, FedEx-LoRA, LoRA-FAIR) in a sharper light than our earlier reading: in this regime their fix is not incidental bookkeeping — it is one of the two independent switches that turn catastrophic erosion off. All three registered seeds are complete and clean, so the pre-registered erosion-persists reading (both minorities below backbone at ≥2/3 seeds) fails unanimously: the registered withdrawal stands on the full sample.

8 THE REPAIR

The historical intervention solves $w^* = \arg\max_{w \in \Delta_K} \min_i (Gw)_i$ on a cosine Gram, then applies w^* to raw trainable-A/B factor states. In seed 42, it reduces capped peak erosion to at most 0.0025, yields within-arm $\Delta_{\text{round}} = +0.071$, and in the saturation cell preserves the entire round-1 gain rather than surrendering it — a +0.091 frozen-anchored swing over n_k (+0.0909 by final scores, robust to the statistic choice); uniform scores slightly higher in both cells. These are ordinary-LoRA intervention results from a dirty source snapshot. They do not instantiate the unit-direction vector certified by the LP. The new FedSpan candidate path instead uses a shared frozen A with row-orthogonal, equal-norm rows ($AA^\top = c^2 I$, $c \approx 0.577$ matched to the initializer scale) and finite nonzero active deltas. Before geometry, each raw PEFT block is scaled by its known $\sigma_\ell = \alpha_\ell / r_\ell$ and the effective blocks are concatenated, so $r_k^2 = \sum_\ell \|\sigma_\ell \Delta B^{\text{raw}}_{k,\ell}\|_F^2$. With unit direction u_k, $G_{ij} = \langle u_i, u_j \rangle$, and declared true effective-update step norm s, it uses $c_k = s w_k^* / (r_k \sqrt{w^{*\top} G w^*})$ and applies the same client coefficient blockwise: $B^{\text{raw}}_{t+1,\ell} = B^{\text{raw}}_{t,\ell} + \sum_k c_k \Delta B^{\text{raw}}_{k,\ell}$ when the mixture norm is nonzero. These delta coefficients need not form a simplex and cannot be passed to direct client-state averaging. The candidate excludes nonfinite/zero/tiny clients, handles empty/singleton active sets, and fails closed with explicit logging on solver, cancellation, reconstruction, or declared coefficient-cap failures. Its finite-game payoff is an optimization diagnostic; it is not a signed nDCG guarantee. The current clone file is a cross-seed correlated-direction proxy, not same-broadcast clone-federation validation.

8.1 Validation of the corrected method

Two campaigns now establish the candidate. The first is an eleven-run attribution grid at seed 42 crossing the parameterisation (ordinary trainable A+B versus shared frozen A) with the aggregation rule, plus a control isolating the row-scale of the frozen factor; every run passed an independent validator that recomputes each round's server aggregate from the persisted client states and refuses any disagreement. Within the frozen-A coordinate — the only comparison the grid licenses — FedSpan finished ahead of uniform in both work regimes (final nDCG@10 0.451 versus 0.435 at full epochs; worst client +0.064 versus +0.044), with the historical raw max–min variant *last*, so the gain is attributable to the corrected geometry and normalisation rather than to max–min as such. The same grid shows the row-scale control moving the score by 0.006, about a third of the method's own margin, which is why cross-scale comparisons are excluded throughout.

The second campaign carries the method to the saturation horizon where the pathology lives, at two seeds (Table 4). The mean advantage over frozen-A uniform replicates closely — +0.0186 and +0.0166 — and its *composition* is the informative part: the gain is concentrated on the two smallest clients (ArguAna +0.053/+0.063, SciFact +0.018/+0.009) while the two largest are approximately neutral, marginally positive at one seed and marginally negative at the other (+0.002/−0.001 and +0.002/−0.004). FedSpan is therefore a *redistribution* rather than a uniform improvement: it does not promise majority gains, and maximising the worst client's alignment never implied it would. The mean is also close to the +0.016 measured independently at the shorter horizon. We note plainly that the aggregate advantage is driven by ArguAna; excluding it, the remaining three clients average approximately zero. Whether geometry yields a structural advantage that does not depend on a single client is precisely what the pre-registered clone federation tests.

The solver behaved as the theory requires, near-identically at both seeds. In all eight rounds of each run the achieved worst-case cosine equalled the optimum of the min-norm program to four decimals (status *optimal*, no fallback rounds), and the protection level declined smoothly from $\gamma^\star = 0.644$ to 0.457 (seed 123) and 0.641 to 0.455 (seed 2024) as clients differentiated — throughout, more than five orders of magnitude above the fail-closed cancellation gate, so the direction-sensitivity regime discussed in §9 was never entered. Every client was still improving at the final round, in contrast to the n_k trajectories that peak at round 1.

Two structural facts sharpen what the direction rule can claim. First, the protection level admits an exact characterization: $\gamma^\star = \operatorname{dist}(0, \operatorname{conv}\{u_k\})$, so $\gamma^\star > 0$ precisely when $0 \notin \operatorname{conv}\{u_k\}$ — no linear-independence or definiteness assumption is needed, and a singular Gram is compatible with large protection. Second, the rule is *clone-invariant by construction*: exact duplicates of a client direction leave the convex hull, hence the applied direction and $\gamma^\star$, unchanged, whereas uniform weighting over clients shifts its aggregate under duplication — the asymmetry the clone-federation experiment tests in the near-duplicate regime.

We state the scope precisely. The within-coordinate margin is a two-seed measurement at this horizon; the shorter-horizon margin is a separate single-seed measurement; and the frozen coordinate itself remains behind ordinary LoRA on raw score at the shorter horizon (they are at parity at R=8), so freezing A is justified by exactness of both aggregation *and application* — and, per the mechanism discriminator above, exactness is not cosmetic here: the frozen coordinate is the one in which n_k's absolute harm does not occur — an arbitrary aggregate step has no exact rank-r factored representation when both factors train — together with auditability and communication, rather than by accuracy.

9 LIMITATIONS

Single backbone family; K=4 proxy silos; the saturation-cell phenomenon holds on three seeds and its weighting-rule control on two, while the within-coordinate FedSpan margin is a two-seed measurement whose aggregate is driven by one client; dirty-code provenance applies to the historical max–min rows only (the attribution grid and all later campaigns run from pinned clean commits under an independent validator); no executed close common-descent baselines yet; no same-broadcast clone federation; and no completed changing-session evidence. The peak-then-erode phenotype is strongest on one client, recall ceilings limit depth analysis on two domains, and in the capped regime erosion is usually forgone gain rather than net harm — the saturation regime

Table 4: FedSpan versus uniform *within* the frozen-A coordinate (R=8, full epochs, two seeds; parameterisation held fixed, so no coordinate confound). Frozen-anchored nDCG@10 drift; gap = FedSpan − uniform. The advantage is a redistribution: it lands on the smallest clients and is approximately neutral on the largest.

Client (train pairs)	gap (s123)	gap (s2024)	mean gap
NFCorpus (110k)	+0.0022	−0.0009	+0.0007
FiQA (14k)	+0.0015	−0.0043	−0.0014
SciFact (0.9k)	+0.0177	+0.0090	+0.0133
ArguAna (0.7k)	**+0.0529**	**+0.0627**	**+0.0578**
mean drift (FedSpan)	+0.1056	+0.1038	+0.1047
mean drift (uniform)	+0.0870	+0.0872	+0.0871
mean gap	**+0.0186**	**+0.0166**	**+0.0176**

is where it becomes absolute. The absolute-erosion phenomenon is coordinate-scoped: it requires joint two-factor training and does not occur under the same weighting in the frozen-A coordinate (all three discriminator seeds clean). And federation's value itself is budget-sensitive: against an equal-budget local-only control, uniform federation wins for some silos and loses for others (SciFact and FiQA train better alone at both completed seeds), so no claim is made that federation dominates non-participation — only that n_k weighting makes it strictly worse than non-participation for every measured silo. We make no differential-privacy, secure-aggregation, Byzantine, partial-participation, cross-device, signed nDCG, or broad optimizer-novelty claim. Threat model, stated explicitly: C1 is a *data-locality* constraint, not a privacy guarantee — the server is trusted (honest), observes individual client adapters, and computes pairwise update geometry, which is structurally incompatible with sum-only secure aggregation; cryptographic protocols for computing the Gram without individual exposure are future work.

Three method-level limitations are stated with their evidence. *Direction sensitivity:* the normalized direction rotates as $O(\varepsilon/\gamma^\star)$ under a perturbation ε of one client direction, so small-$\gamma^\star$ rounds could apply a full-length step in a noise-dominated direction; in the measured runs $\gamma_t^\star$ remained in $[0.457, 0.644]$ — more than five orders of magnitude above the fail-closed gate — so this regime was not entered, and a soft-brake step policy is pre-registered as a contingent ablation should it appear. *Stochastic weight bias:* the mixture is solved on directions produced by stochastic multi-step local training and is therefore a biased estimate of the population-level protective mixture [6], on top of the delta-versus-gradient gap; a server-side correction in the MoCo style is future work. *First-order license:* treating local update deltas as gradient proxies is the standard first-order license carried by the entire GEM/MGDA lineage [17, 21]; our alignment-to-next-round-score correlations are measured, not assumed, and the descent statement is confined to the exact-gradient corollary above.

10 CONCLUSION

The three-seed capped cells show a repeatable ranking-level warning: size-proportional aggregation forfeits gains that uniform federation provides, first near the top ranks. At the saturation horizon the warning becomes absolute and replicates on three seeds: minority clients finish below the untrained backbone under n_k (SciFact -0.074 ± 0.004, ArguAna -0.019 ± 0.005) while a uniform control at the same seed lifts every client. The pre-registered discriminator attributes the harm to an *interaction*: the same n_k weighting in a frozen-A coordinate eliminates the absolute erosion at both completed seeds, so skew is necessary but not sufficient — skew acting through joint two-factor training is what destroys. The corrected frozen-span method passed an eleven-run attribution grid under an independent validator and, at the saturation horizon on two seeds, redistributes toward the smallest clients (mean advantage +0.018, concentrated on the two minorities, approximately neutral on the two majorities), its solver provably optimal in every round. What remains before claiming a general cure is exactly the pre-registered remainder: the second seed on the corrected method, faithful loss- and common-descent baselines, the same-broadcast clone federation where the geometry argument must earn its keep, and changing-session evidence.

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